

Exponential family

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Para Int.

Suppose X has pmf/pdf $f(x|\theta); \theta \in \Theta$. If $f(x|\theta)$ can be written as

$$f(x|\theta) = h(x) e^{\sum_{i=1}^k \eta_i(\theta) T_i(x) - B(\theta)}, \quad k \in \mathbb{N}$$

for some real valued fns. $\eta_i(\cdot), T_i(\cdot), B(\cdot)$, where

$\eta_i(\cdot)$ and $B(\cdot)$ are functions of θ only (no x involved)
and $T(\cdot)$ and $h(\cdot)$ ~ ~ ~ ~ ~ x ~ ~ ~ ~ ~ $(\theta \sim)$

Alt $f(x|\theta) = h(x) c(\theta) e^{\sum \eta_i(\theta) T_i(x)}$

Ex $f(x|\theta) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x^2 - 2x\theta + \theta^2)}$

$$= \underbrace{\left(\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \right)}_{h(x)} e^{x\theta - \frac{\theta^2}{2}}$$

$$\begin{matrix} h(x) & T(x) = x, & \eta(\theta) = \theta \\ k=1 \end{matrix}$$

Ex $\mathcal{N}(\mu, \sigma^2) \quad \theta = (\mu, \sigma^2)$

$$f(x|\theta) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x^2 - 2\mu x + \mu^2)}$$

$$= \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{x^2}{2\sigma^2} + \frac{\mu}{\sigma^2} x - \frac{\mu^2}{2\sigma^2} - \frac{1}{2} \ln \sigma^2 \right\}$$

$\sim$
 $h(x)$

$$T_1(x) = x^2, \eta_1(\theta) = -\frac{1}{2\sigma^2} \quad B(\theta) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2} \ln \sigma^2$$

$$T_2(x) = x, \eta_2(\theta) = \frac{\mu}{\sigma^2}$$

Idea: To be able to write $f(x|\theta)$ as products of fns. that involve only x or only θ .

Ex. $X \sim U(0, \theta); \theta > 0$ $f(x|\theta) = \frac{1}{\theta} \mathbb{1}_{(0 < x < \theta)}$
 $\notin$ exponential family.

Note: Support $\mathcal{X} = \{x: f(x|\theta) > 0\}$ does not depend on θ when $f(x|\theta)$ belongs to exponential family.

Ex. $X \sim \text{Bin}(n, \theta)$; $0 < \theta < 1$ (n known)

$$\begin{aligned} f(x|\theta) &= \binom{n}{x} \theta^x (1-\theta)^{n-x} ; x=0, 1, \dots, n \\ &= \underbrace{\binom{n}{x}}_{h(x)} \exp \left\{ \underbrace{x \ln \theta}_{T_1(x) \eta_1(\theta)} + \underbrace{(n-x) \ln (1-\theta)}_{T_2(x) \eta_2(\theta)} \right\} \\ &= \underbrace{\binom{n}{x}}_{h(x)} \exp \left\{ \underbrace{x \ln \left(\frac{\theta}{1-\theta} \right)}_{T(x) \eta(\theta)} + \underbrace{n \ln (1-\theta)}_{B(\theta)} \right\} \end{aligned}$$

To make the notion of "k meaningful" we try to have a "minimal representation" i.e. where $T_1(x), \dots, T_k(x)$ do not satisfy a linear constraint and $\eta_1(\theta), \dots, \eta_k(\theta)$ " " " " " " " .

Defⁿ: $f(x|\theta)$ belongs to a k -parameter exponential family if
 $f(x|\theta) = h(x) e^{\sum_{i=1}^k \eta_i(\theta) T_i(x) - B(\theta)}$ and the representation
 is minimal.

- Ex.
- 1) $\text{Bin}(n, \theta) \in$ 1-parameter exp family
 - 2) $\text{Poi}(\lambda) \in$ " "
 - 3) $\mathcal{N}(\mu, \sigma^2) \in$ 2-parameter "
 - 4) $\text{Gamma}(\alpha, \beta) \in$ "
 - 5) $\text{Beta}(\alpha, \beta) \in$ "

Not exponential

- 1) $f(x|\mu, \sigma) = \frac{1}{\sigma \pi (1 + (\frac{x-\mu}{\sigma})^2)}$
- 2) $f(x|\mu, \sigma) = \frac{1}{2\sigma} e^{-|\frac{x-\mu}{\sigma}|}$

HW Show that

$f(x|\mu) = \frac{1}{2} e^{-|x-\mu|} \notin$ 1-parameter exponential family

To help understand "k" better, let's reparametrize the family

$$f(x|\theta) \text{ as } f(x|\eta) = \binom{n}{x} \exp\{x\eta - A(\eta)\}$$

$$\eta = \ln\left(\frac{\theta}{1-\theta}\right)$$

$$A(\eta) = n \ln(1 + e^\eta)$$

Def. $f(x|\eta) = h(x) e^{\sum_{i=1}^k \eta_i T_i(x) - A(\eta)}$
 is said to be in "canonical form".

where $\int f(x|\eta) = 1 \Leftrightarrow \int h(x) e^{\sum_{i=1}^k \eta_i T_i(x)} = e^{A(\eta)} < \infty$

The set of possible values of $\eta = (\eta_1, \dots, \eta_k)$ is called
 the natural parameter space $\mathcal{N} := \left\{ \eta : \int h(x) e^{\sum_{i=1}^k \eta_i T_i(x)} < \infty \right\}$
 $\eta = \text{natural parameter}$

* The Natural Parameter Space is convex

Take any two points $\eta^{(1)}$ and $\eta^{(2)} \in \mathcal{N}$

Show $\alpha \eta^{(1)} + (1-\alpha) \eta^{(2)} \in \mathcal{N}$ (H.W.)

Hint: Use Hölder's inequality.

Ex. $N(\mu, \sigma^2)$ $\theta = (\mu, \sigma^2)$

$$f(x|\theta) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{x^2}{2\theta^2} + \frac{\mu x}{\sigma^2} - \frac{\mu}{2\sigma^2} - \frac{1}{2} \ln \sigma^2 \right\}$$

$$\eta_1(\theta) = -\frac{1}{2\sigma^2}$$

$$\eta_2(\theta) = \frac{\mu}{\sigma^2} \quad \mathcal{N} := \left\{ (\eta_1(\theta), \eta_2(\theta)) : \mu \in \mathbb{R}, \sigma > 0 \right\}$$

$$= \mathbb{R} \times \mathbb{R}$$

Sufficiency in exponential family:

Suppose $X = (X_1, \dots, X_n)$ has pmf/pdf $f(x|\theta)$
 $= h(x) e^{\sum_{i=1}^k \eta_i(\theta) T_i(x) - B(\theta)}$

Then $T(X) = (T_1(X), \dots, T_k(X))$ is sufficient for θ by NFFT

is also sufficient for the natural parameter

$$\eta = (\eta_1(\theta), \dots, \eta_k(\theta))$$

If $X_i \text{ iid}$ $f(x|\theta) = h(x) e^{\sum_j \eta_j(\theta) T_j(x) - B(\theta)}$

Then $(X_1, \dots, X_n) \sim f(x|\theta) = \prod_i h(x_i) e^{\sum_j \eta_j(\theta) \sum_i T_j(x_i) - nB(\theta)}$

Define $T_j(\underline{x}) = \sum_{i=1}^n T_j(x_i)$ then $T(\underline{x}) = (T_1(\underline{x}), \dots, T_k(\underline{x}))$
 is sufficient for θ (as well as for η)

Qn: When $T(\underline{x}) = (T_1(\underline{x}), \dots, T_k(\underline{x}))$ is minimal suff

want $\frac{f(x|\eta)}{f(y|\eta)}$ is free of $\eta \Leftrightarrow T(x) = T(y)$

$$\frac{h(x) e^{\sum_{j=1}^k \eta_j T_j(x) - A(\eta)}}{h(y) e^{\sum_{j=1}^k \eta_j T_j(y) - A(\eta)}} = \frac{h(x)}{h(y)} e^{\sum_{j=1}^k \eta_j (T_j(x) - T_j(y))}$$

is free of η

$$= \frac{h(x)}{h(y)} e^{\eta'(T(x) - T(y))} \quad \eta = (\eta_1, \dots, \eta_k)$$

$$T(x) = (T_1(x), \dots, T_k(x))$$

Need some restriction on $\mathcal{N}$

Let's consider $(k+1)$ points, say $\eta^{(0)}, \eta^{(1)}, \dots, \eta^{(k)}$ from $\mathcal{N}$

and $\mathcal{P} = \{f(x|\eta) : \eta \in \mathcal{N}\}$ and subfamily

$$\mathcal{P}_0 = \{f(x|\eta^{(j)}) : j=0, 1, \dots, k\}$$

Enough to check when

$T(\underline{x}) = (T_1(\underline{x}), \dots, T_k(\underline{x}))$ is min_suff for $\mathcal{P}_0$.

We know, $S(\underline{x}) = \left(\frac{f(x|\eta^{(1)})}{f(x|\eta^{(0)})}, \dots, \frac{f(x|\eta^{(k)})}{f(x|\eta^{(0)})} \right)$ is a min_suff
stat for $\mathcal{P}_0$

If $S(x) = S(y) \Leftrightarrow T(x) = T(y)$ then T is min_suff
as well.

$$\text{Consider } \frac{f(x|\eta^{(j)})}{f(x|\eta^{(0)})} = e^{\sum_{i=1}^k (\eta_i^{(j)} - \eta_i^{(0)}) T_i(x)}$$

$$= e^{(\eta^{(j)} - \eta^{(0)})' T(x)}$$

$S(x)$ is equivalent to $(\eta^{(1)} - \eta^{(0)})' T(x), \dots, (\eta^{(k)} - \eta^{(0)})' T(x)$

Q. When $\begin{bmatrix} (\eta^{(1)} - \eta^{(0)})' T(x) \\ \vdots \\ (\eta^{(k)} - \eta^{(0)})' T(x) \end{bmatrix} = M^{k \times k} T(x)$ is equivalent to $T(x)$?

So, $T(x)$ is min suff for η if M is non-singular

Sufficient conditions

- 1) If $\mathcal{N}$ contains a k -dim. rectangle in $\mathbb{R}^k$
- 2) $\exists$ points $\eta^{(0)}, \dots, \eta^{(k)}$ that span $\mathcal{N}$ (weaker condition)

Defⁿ: $f(x|\theta) = h(x) \exp \left\{ \sum_{i=1}^k \eta_i(\theta) T_i(x) - B(\theta) \right\}$

(or equiv., $f(x|\eta) = h(x) \exp \left\{ \sum_{i=1}^k \eta_i T_i(x) - A(\eta) \right\}$)

is called a "full rank" k -parameter exponential family if the above is a minimal representation and the natural parameter space $\mathcal{N}$ contains a k -dimensional rectangle in $\mathbb{R}^k$.

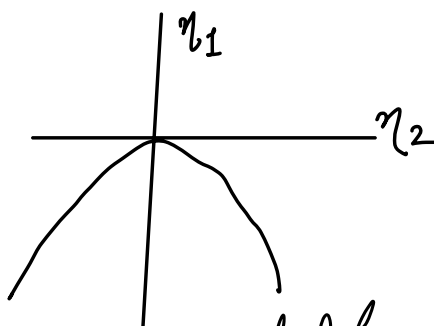
Ex. $\mathcal{N}(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$

$$\mathcal{N} = \left\{ \left(-\frac{1}{2\sigma^2}, \frac{\mu}{\sigma^2} \right) : \mu \in \mathbb{R}, \sigma > 0 \right\} = \mathbb{R} \times \mathbb{R}^+$$

$$\begin{aligned}
 \mathcal{N}(\theta, \theta^2) \quad f(x|\theta) &= \frac{1}{|\theta| \sqrt{2\pi}} e^{-\frac{1}{2\theta^2}(x^2 - 2x\theta + \theta^2)} \\
 &= \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{x^2}{2\theta^2} + \frac{x}{\theta} - \frac{1}{2} - \ln|\theta| \right\}
 \end{aligned}$$

$$\mathcal{N} = \left\{ \left(-\frac{1}{2\theta^2}, \frac{1}{\theta} \right) : \theta \in \mathbb{R} \right\}$$

$$\begin{aligned}
 \eta_1(\theta) &= -\frac{1}{2\theta^2} & \eta_1(\theta) &= -\frac{1}{2}\eta_2^2(\theta) \\
 \eta_2(\theta) &= \frac{1}{\theta}
 \end{aligned}$$



$\mathcal{N}(\theta, \theta^2) \notin$ a full rank exponential family
 Such a family is called a "curved exponential family"

Other ex. $f(x|\theta) = ce^{-(x-\theta)^4}, \mathcal{N}(\theta, \theta^4)$

Thm: Suppose $X = (X_1, \dots, X_n)$ has pmf/pdf $f(x|\theta) = h(x)e^{\sum_1^k \eta_i(\theta)T_i(x) - B(\theta)}$ that belongs to a k -parameter full rank exponential family then $T(X) = (T_1(X), \dots, T_k(X))$ is a complete and sufficient statistic for θ